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MODEL SELECTION AND VALIDATION REPORT ( general_teve model )
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## OBSERVATIONS:3451

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( general_teve model )

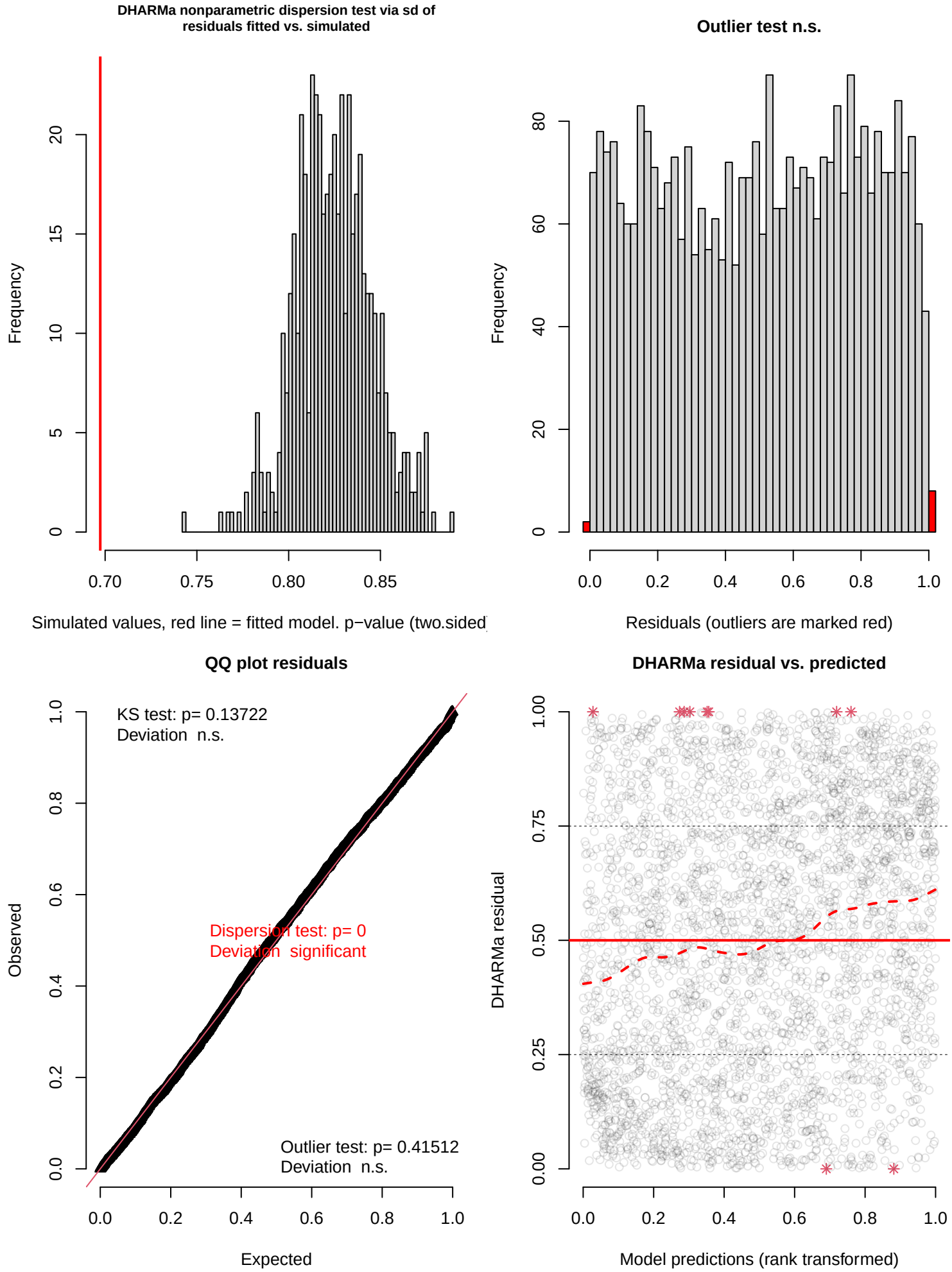
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

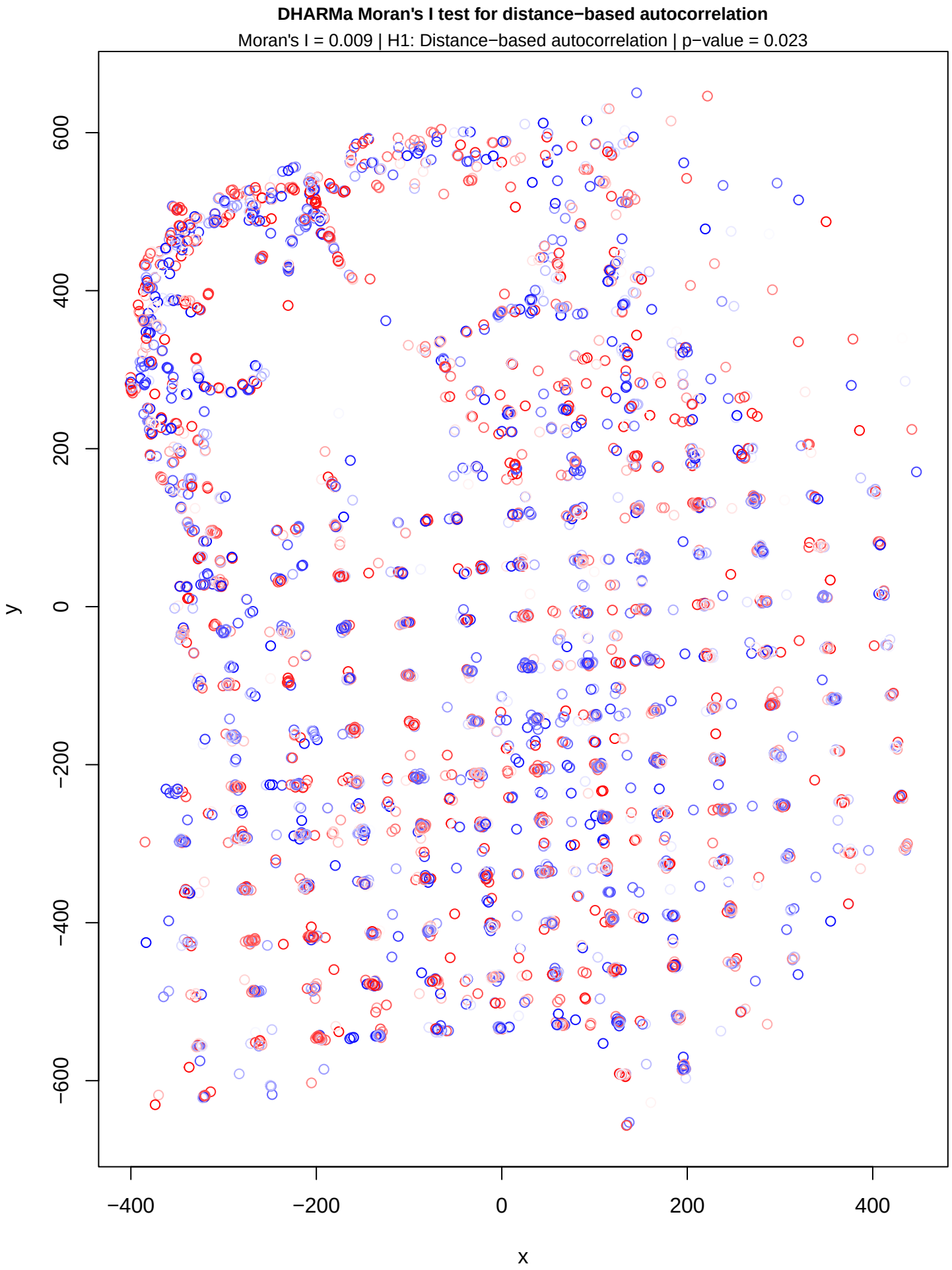
Conditional model:
              coef.est coef.se
(Intercept)      −0.05    0.09
sbt_mean          0.07    0.05
sbt_sd           −0.04    0.03
sst_sd           −0.01    0.03
sic_mean         −0.09    0.04
log_chla_mean     0.00    0.02
depth             0.00    0.03

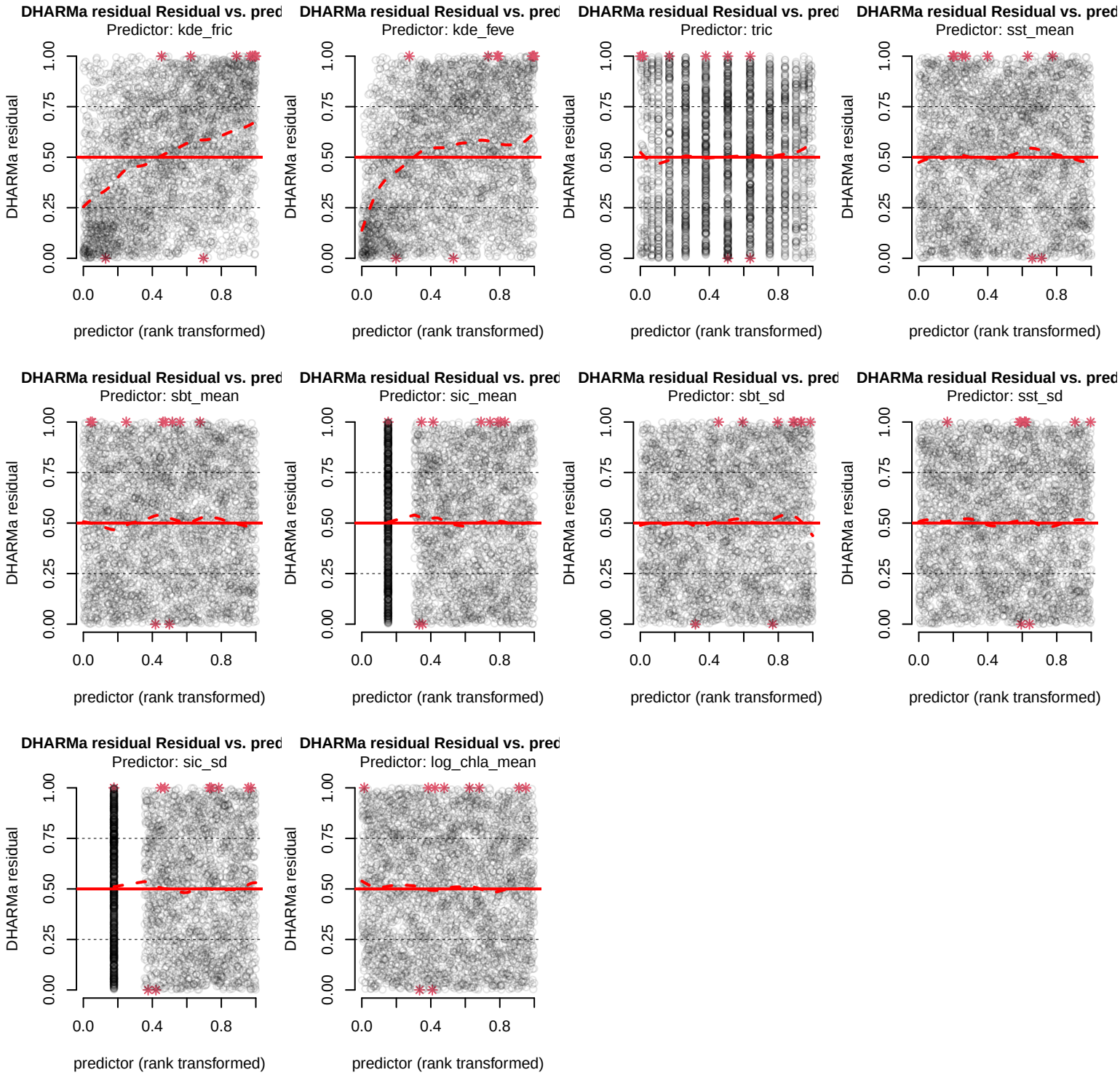
Dispersion parameter: 0.88
Spatiotemporal AR1 correlation (rho): 0.87
Matérn range: 216.64
Spatiotemporal marginal AR1 SD: 0.52
REML criterion at convergence: 4678.383

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                   11 9378.767   3.657991
spatiotemporal_random_fields_ar1_with_anisotropy 13 9382.425 127.736217
spatial_random_fields                              10 9510.161   3.516571
spatial_random_fields_with_anisotropy              12 9513.677   7.338357
spatiotemporal_random_fields_iid                   10 9521.016   3.331714
spatiotemporal_random_fields_iid_with_anisotropy 12 9524.348 184.378369
error_only                                          8 9708.726         NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
20
0.70
0.75
0.80
0.85
Simulated values, red line = fitted model. p−value (two.sided
Outlier test n.s.
Frequency
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.13722
Deviation n.s.
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0.41512
Deviation n.s.
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected
DHARMa residual vs. predicted
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.009 | H1: Distance−based autocorrelation | p−value = 0.023
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: kde_fric
Predictor: kde_feve
Predictor: tric
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8